THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 7 — MATHS · NCERT-TYPE FULL-YEAR PAPER SET 2

Full Syllabus (Chapters 1–8) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. Two parallel lines are cut by a transversal. An interior angle is 65°. The interior angle on the same side is: 1

(a) 115° **(b)** 125° **(c)** 105° **(d)** 135°

2. The number of steps the Collatz sequence from 5 takes to reach 1 is: 1

(a) 5 **(b)** 4 **(c)** 6 **(d)** 7

3. The exterior angle of a triangle is $(3x + 10)^\circ$ and the interior opposite angles are 50° and 50°. The value of x is: 1

(a) 30 **(b)** 29 **(c)** 32 **(d)** 31

4. The number of $\frac{1}{4}$s in 2 is: 1

(a) 8 **(b)** 6 **(c)** 2 **(d)** 4

5. The place value of the digit 4 in 34,56,789 (the leftmost 4) is: 1

(a) 40,000 **(b)** 4,00,000 **(c)** 4 **(d)** 40,00,000

6. Insert brackets to make true: $3 + 4 \times 5 - 6 = 29$. The correct form is: 1

(a) $(3 + 4 \times 5) - 6$ **(b)** $3 + (4 \times 5) - 6$ **(c)** $3 + 4 \times (5 - 6)$ **(d)** $(3 + 4) \times 5 - 6$

7. 2.5 kg expressed in grams is: 1

(a) 25000 **(b)** 2500 **(c)** 250000 **(d)** 250

8. Subtract $4x$ from $9x$. The result is: 1

(a) $-5x$ **(b)** $13x$ **(c)** 5 **(d)** $5x$

9. If a transversal makes equal corresponding angles with two lines, the lines are: 1

(a) coincident **(b)** parallel **(c)** intersecting **(d)** perpendicular

10. The smallest digit $*$ making $4 * 3$ divisible by 9 is: 1

(a) 2 **(b)** 1 **(c)** 3 **(d)** 4

11. If $\triangle ABC \cong \triangle PQR$, then side BC equals: 1

(a) PQ **(b)** AB **(c)** PR **(d)** QR

12. $1\frac{1}{2} \times 4$ equals: 1

(a) $4\frac{1}{2}$ **(b)** 6 **(c)** 3 **(d)** 12

13. The number of zeros in 1,00,000 is: 1

(a) 4 **(b)** 6 **(c)** 5 **(d)** 7

14. Which value is the greater: $6 \times 7 - 5$ or $5 \times 8 - 2$? The larger value is: 1

(a) 39 **(b)** 37 **(c)** 40 **(d)** 38

15. ₹7.25 expressed in paise is: 1

(a) 72500 **(b)** 7250 **(c)** 725 **(d)** 72.5

16. The value of $2(x + 3)$ for $x = 4$ is: 1

(a) 15 **(b)** 14 **(c)** 16 **(d)** 13

17. Angles A and B are vertically opposite and $A = 28°$. Then $B =$ 1

(a) $48°$ **(b)** $18°$ **(c)** $28°$ **(d)** $38°$

18. The sum $1 + 2 + 3 + \cdots + 10$ is: 1

(a) 55 **(b)** 54 **(c)** 56 **(d)** 57

19. In $\triangle ABC$, the angle included between sides AB and AC is: 1

(a) $\angle A$ **(b)** $\angle A + \angle B$ **(c)** $\angle B$ **(d)** $\angle C$

20. The reciprocal of $1\frac{1}{2}$ (as a fraction) is: 1

(a) $\frac{1}{3}$ **(b)** $\frac{11}{12}$ **(c)** $\frac{2}{3}$ **(d)** $1\frac{1}{3}$

SECTION — B (Very Short Answer Type) — 6 × 2 = 12

Answer in about 20–30 words each.

21. Two sides of a triangle are 6 cm and 10 cm. Find the least and the greatest whole-number length of the third side. 2

22. How many $\frac{1}{8}$s are there in $\frac{3}{4}$? 2

23. What number is 99,999 + 1? What is the successor of the largest 5-digit number? 2

24. Write '12 more than the product of 7 and 9' as an expression and find its value. 2

25. Find $3.6 \div 0.4$. 2

26. Simplify: $7x + 4 - 3x + 6$. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

27. A bus travels 1,25,000 km every year. How many kilometres does it travel in 8 years? Write in words and in the international system. 3

28. A tailor makes 12 shirts, using 3 metres of cloth for each, and has 60 metres of cloth. Write an expression for the cloth left and evaluate it. If cloth costs ₹45 per metre, find the cost of cloth used. 3

29. Express $\frac{3}{8}$ as a decimal and say whether it is greater than or less than 0.4. 3

30. Add $5x + 9y$ and $3x - 4y$. 3

31. A transversal cuts two lines and makes corresponding angles of $65°$ and $65°$. Are the lines parallel? Give a reason. 3

OR

Find how many reverse-and-add steps turn 57 into a palindrome, showing each step. 3

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32. In $\triangle ABC$ and $\triangle PQR$, $AB = PQ = 5$ cm, $BC = QR = 7$ cm and $\angle B = \angle Q = 50^\circ$. Name the congruence rule, and find PR if $AC = 9$ cm. Find also the perimeter of $\triangle PQR$. 3

OR

Divide $2\frac{1}{2}$ by $\frac{3}{4}$ and write the answer as a mixed number. 3

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. Find the value of $12 + (18 - 6) \times 4 \div 8 - 5$. Explain the order in which you did the operations. 5

34. A runner covers 2.5 km, 3.25 km and 2.75 km in three laps. Find the total distance, how much farther the longest lap is than the shortest, and how much more she must run to complete 10 km. 5

35. A hall has 8 rows of r chairs and 6 extra chairs at the back. Write an expression for the total number of chairs and find it for $r = 12$ and for $r = 14$. 5

36. An angle is 2 times its supplement. Find the angle and its supplement. 5

OR

Write the Collatz sequence from 17. Count how many times the number is halved and how many times it is tripled and increased by 1. 5

37. Two sides of a triangle are 7 cm and 12 cm. 5
(i) Between which two lengths does the third side lie?
(ii) How many whole-number lengths are possible?
(iii) Find the smallest and the largest possible perimeter in whole centimetres.

OR

A garden is $\frac{4}{5}$ hm long and $\frac{5}{6}$ hm wide. $\frac{3}{4}$ of it is used for vegetables and the rest for flowers. Find the area of the garden, of the vegetable part and of the flower part (in hm^2). 5

38. A government school uses 2,400 litres of water per day. Find the water used in a year of 365 days, round it to the nearest lakh and write the rounded value in words. 5

OR

A fruit seller buys 25 kg of apples at ₹60 per kg and 10 kg of mangoes at ₹150 per kg. He sells all the fruit for ₹3200. Write an expression for his profit or loss and find it. 5

END OF QUESTION PAPER — **Total: 80 Marks**

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